بسم الله الرحمن الرحيم

Prof. Moha

Amino Acids Metabolism

Ibrahim Selim

الأستاذ الكتور

محمد إبراهيم سليم

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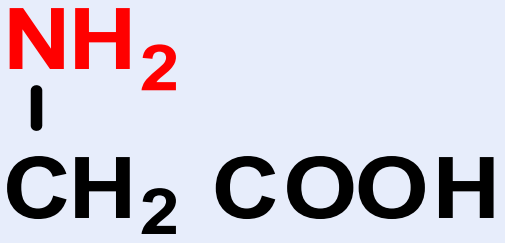
Lecture 2

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•Glycine Metabolism

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Glycine



Structure

- It is a-amino acetic acid

Nutrional Value

- It is non-essential amino acid

Metabolic Fate

- It is glucogenic amino acid

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Glycine

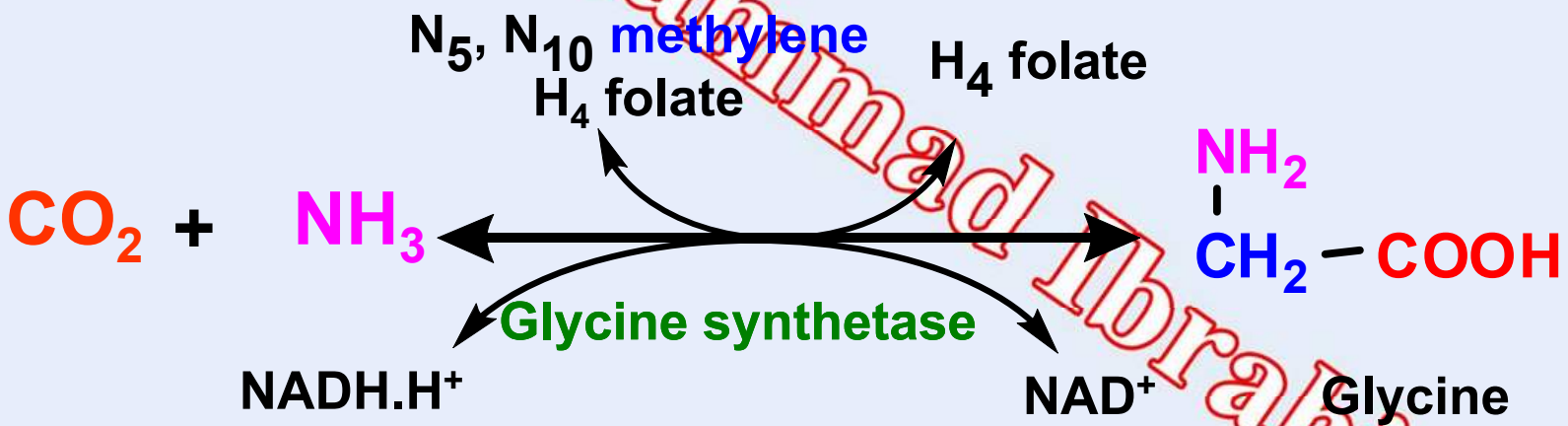
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Biosynthesis

1. From CO_2 and NH_3
2. From glyoxylic acid
3. From serine
4. From threonine
5. From choline
6. From hydroxy proline

Glycine

1- Biosynthesis from CO_2 and NH_3

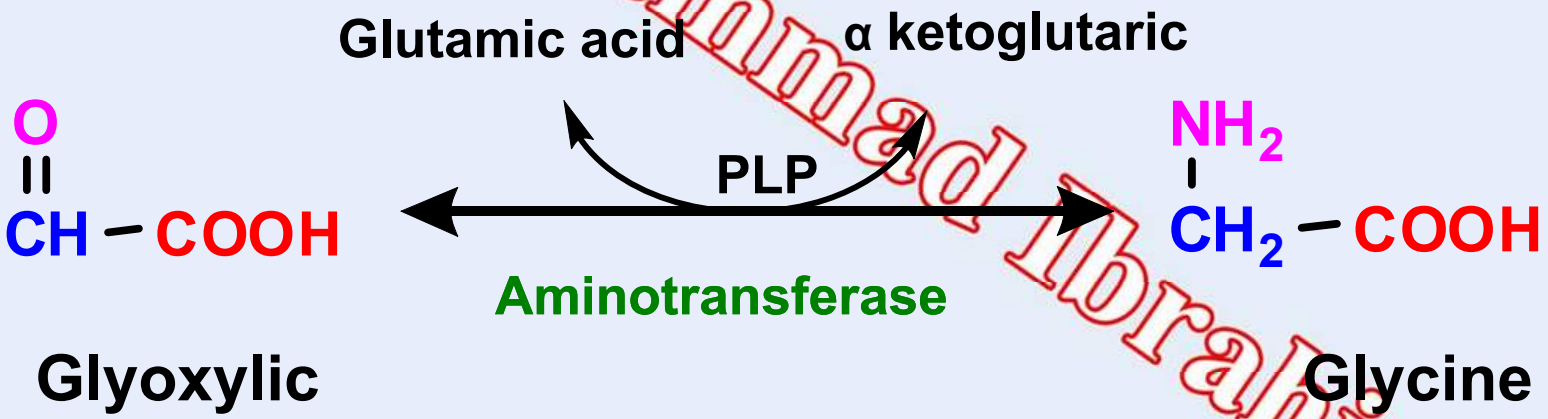


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Glycine

2- Biosynthesis from glyoxylic acid

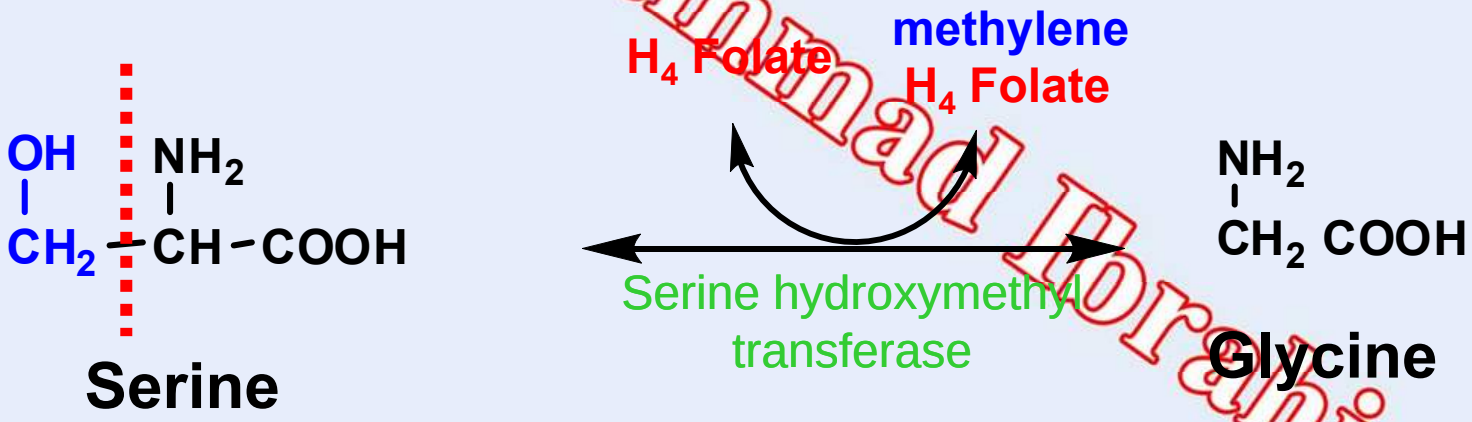


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Glycine

3- Biosynthesis from serine



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Glycine

4- Biosynthesis from threonine

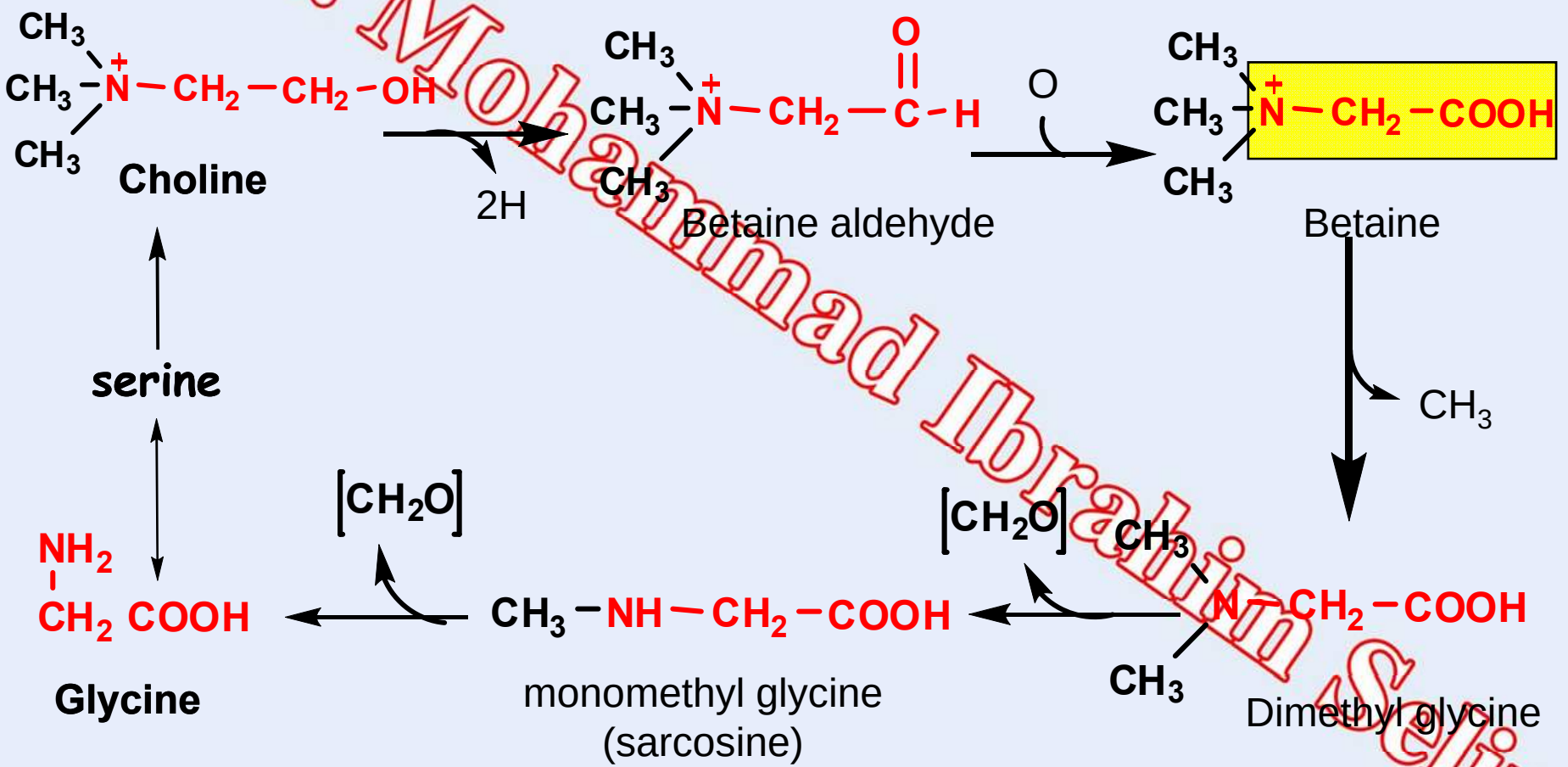


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Glycine

5- Biosynthesis from choline

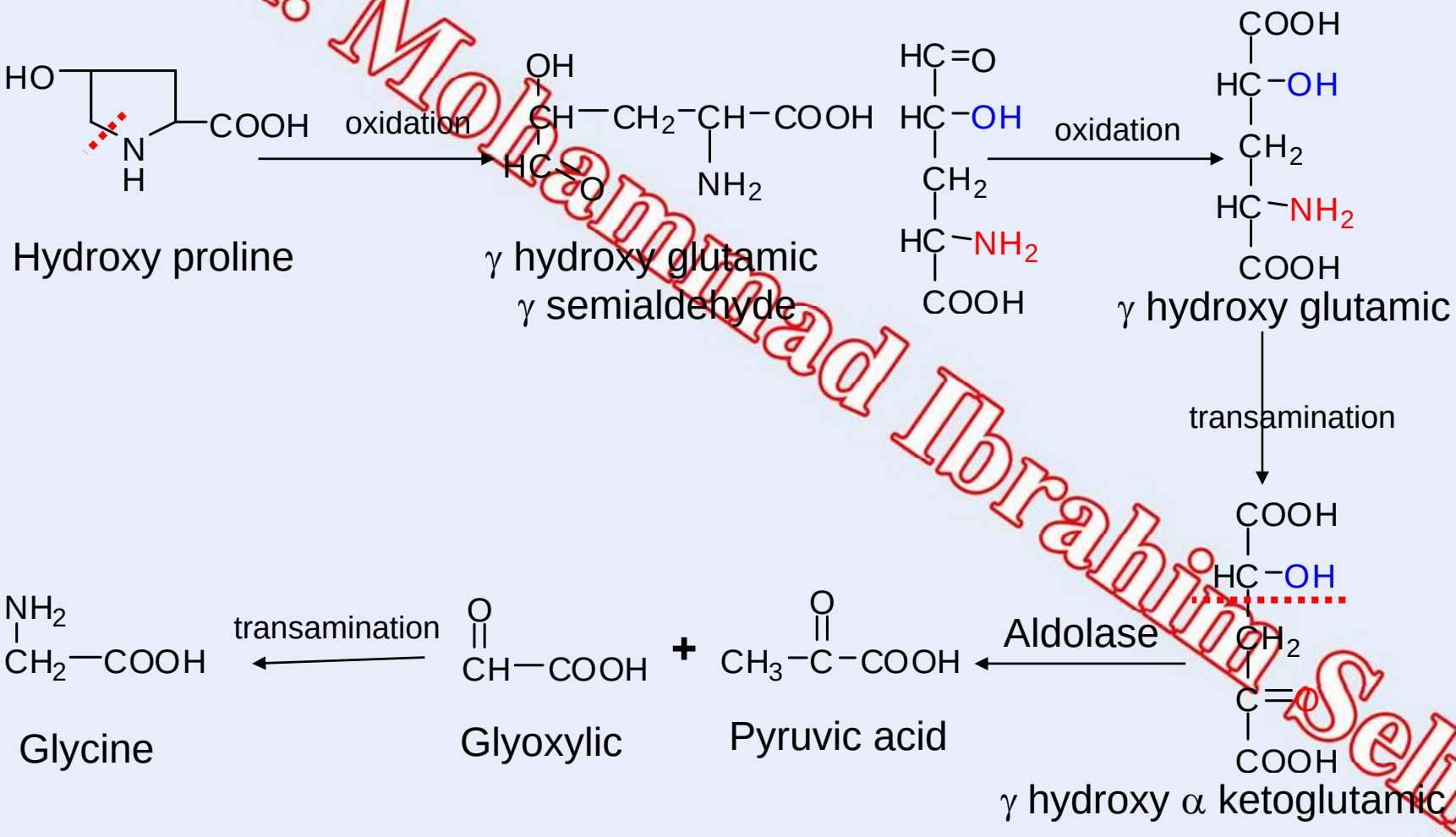


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Glycine

6- Biosynthesis from hydroxy proline



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Glycine

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Biosynthesis

1. From CO_2 and NH_3
2. From glyoxylic acid
3. From serine
4. From threonine
5. From choline
6. From hydroxy proline

Glycine

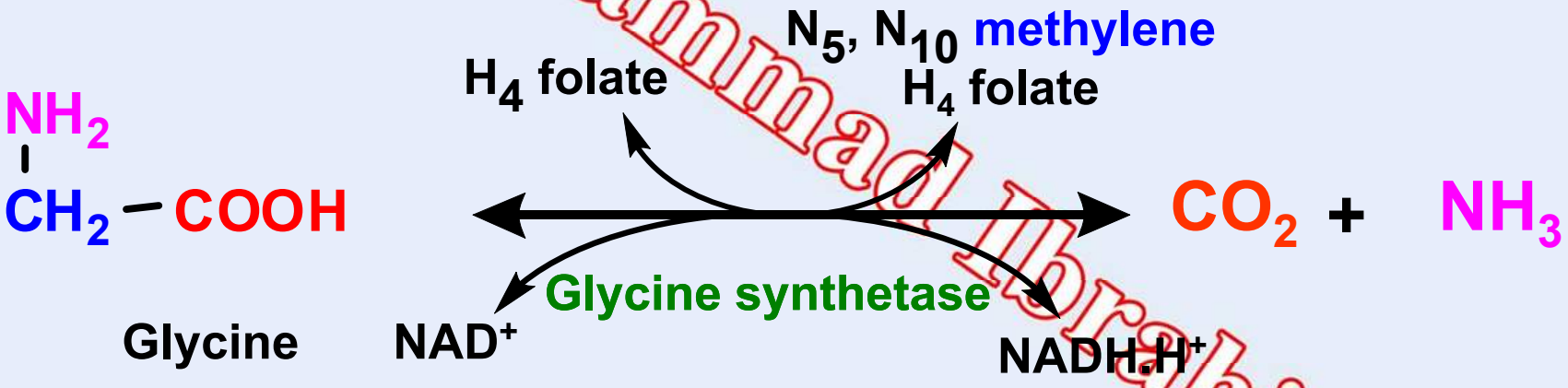
Metabolic Fate

1. Cleavage to CO_2 and NH_3
2. Formation of glyoxylic acid
3. Biosynthesis of serine
4. Synthesis of glucose
5. Creatine and creatinrne synthesis
6. Synthesis of glutathione
7. Haeme biosynthesis
8. Synthesis of purine bases
9. Detoxication of benzoic acid
10. Conjugation with bile acids
11. Incorporated in tissue proteins

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Glycine

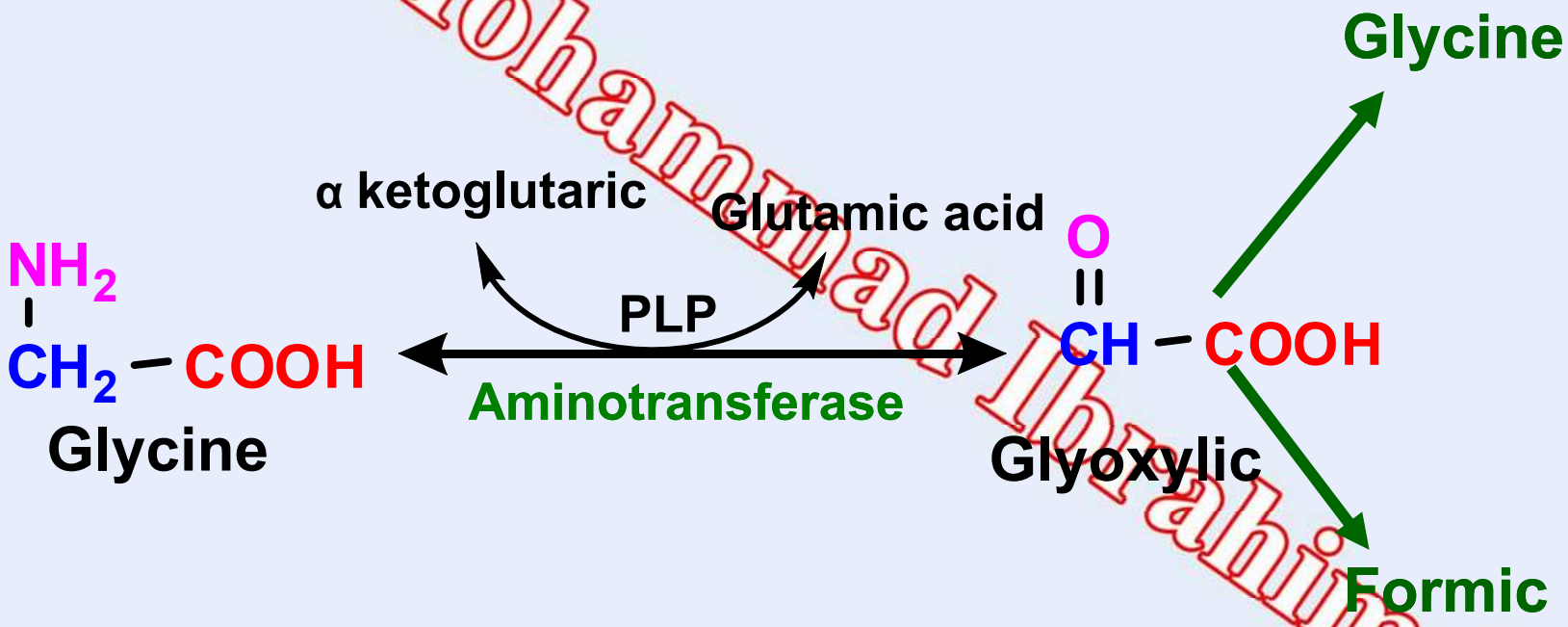
1- Cleavage into CO_2 and NH_3



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Glycine

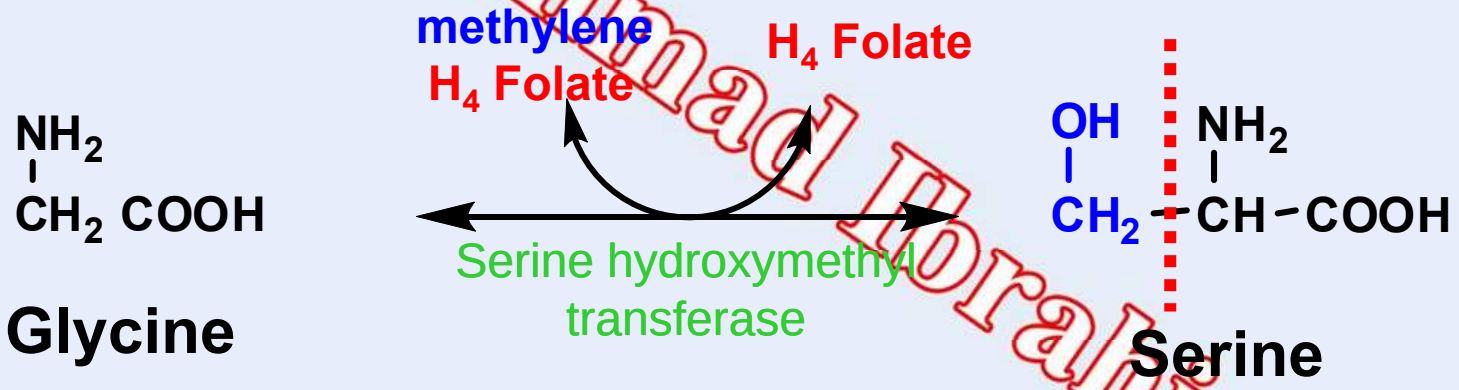
2- Formation of glyoxylic acid



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Glycine

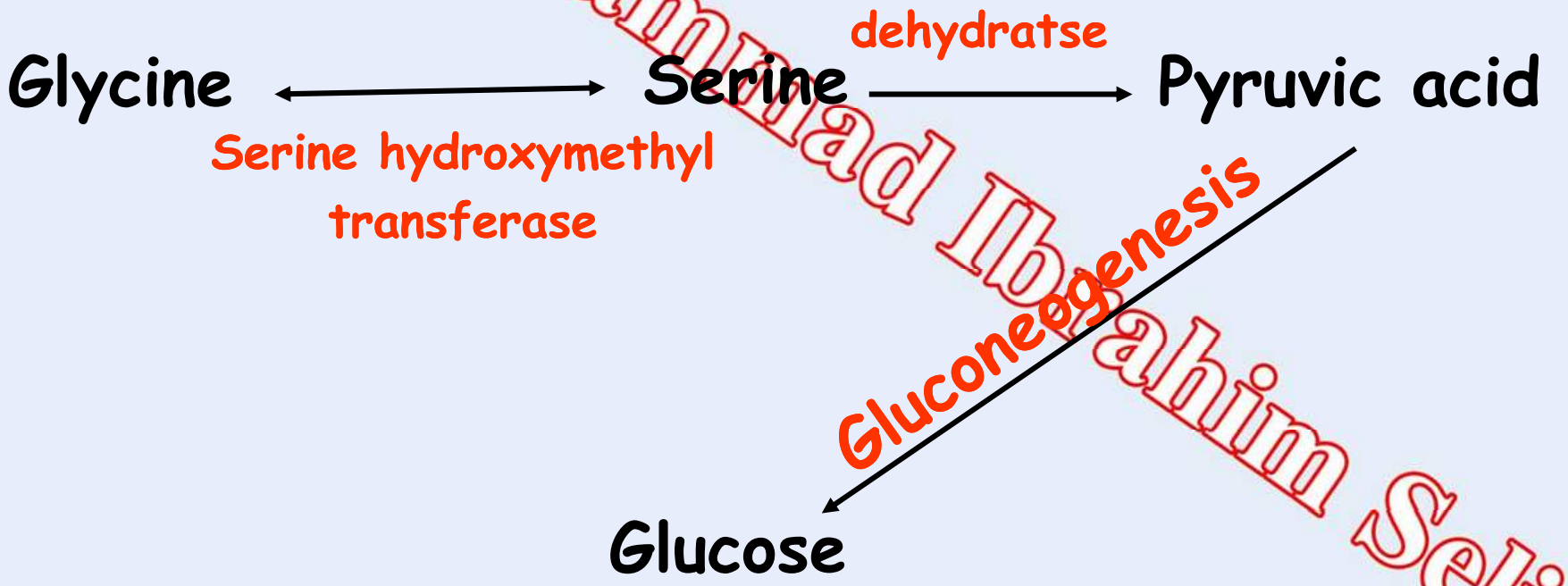
3- Biosynthesis of serine



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Glycine

4- Glucose synthesis

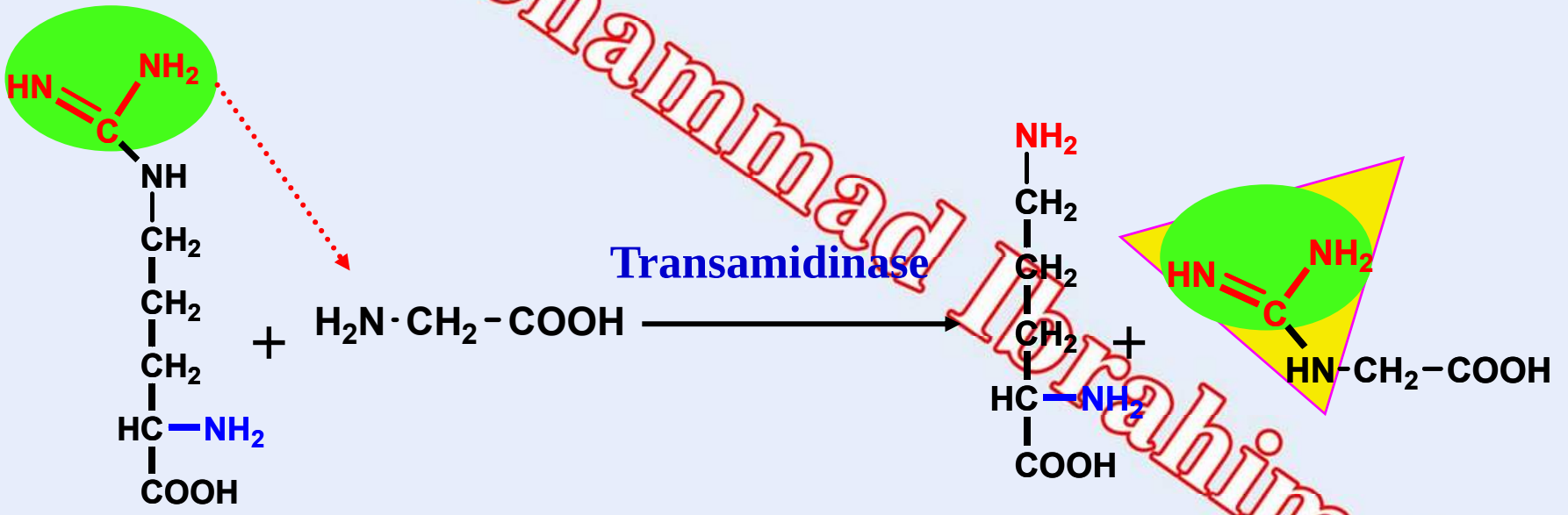


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Glycine

5- Biosynthesis of Creatine and creatinrne

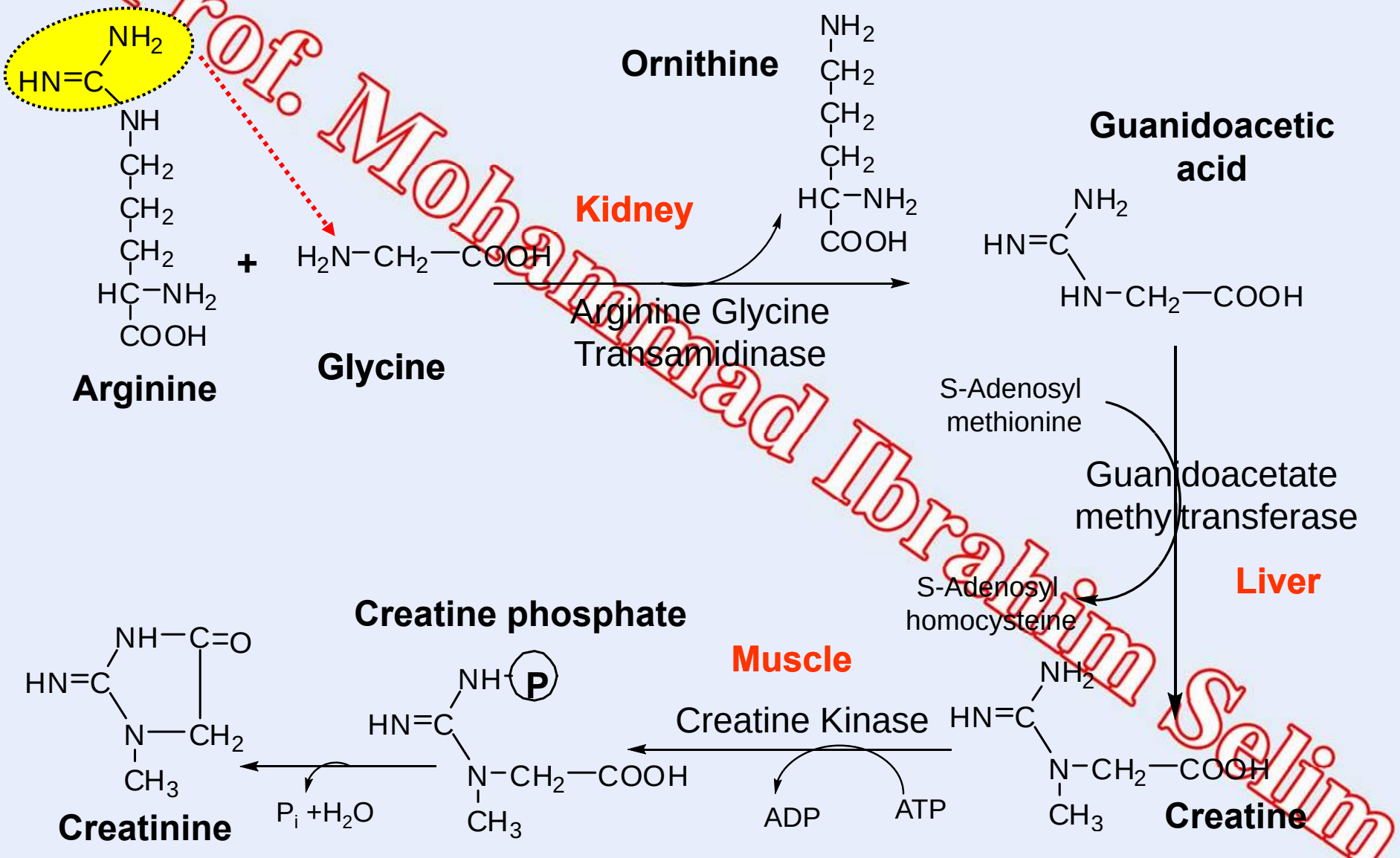
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Arginine **Glycine** **Ornithine** **Guanidoacetic acid**

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Biosynthesis of Creatine and creatinine



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Creatine and Creatinine

- **Blood level**
  - Creatine 0.2-0.6 mg/dl.
  - Creatinine 0.5-1.2 mg/dl
- **Urinary excretion**
  - Normally creatine is present in very small amount in urine.
  - Normal urine creatinine levels: 1 – 2 g/24 hours
  - Creatinine excretion for each individual is constant
- **Creatinine coefficient**
  - Males: 20 – 30 mg/kg/24 hours
  - Females: 10 – 20 mg/kg/ 24 hours
- **Creatinine Clearance**
  - Males: 90 - 130 ml/minute
  - Females: 80 – 120 ml/minute

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Creatinuria

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Physiological causes

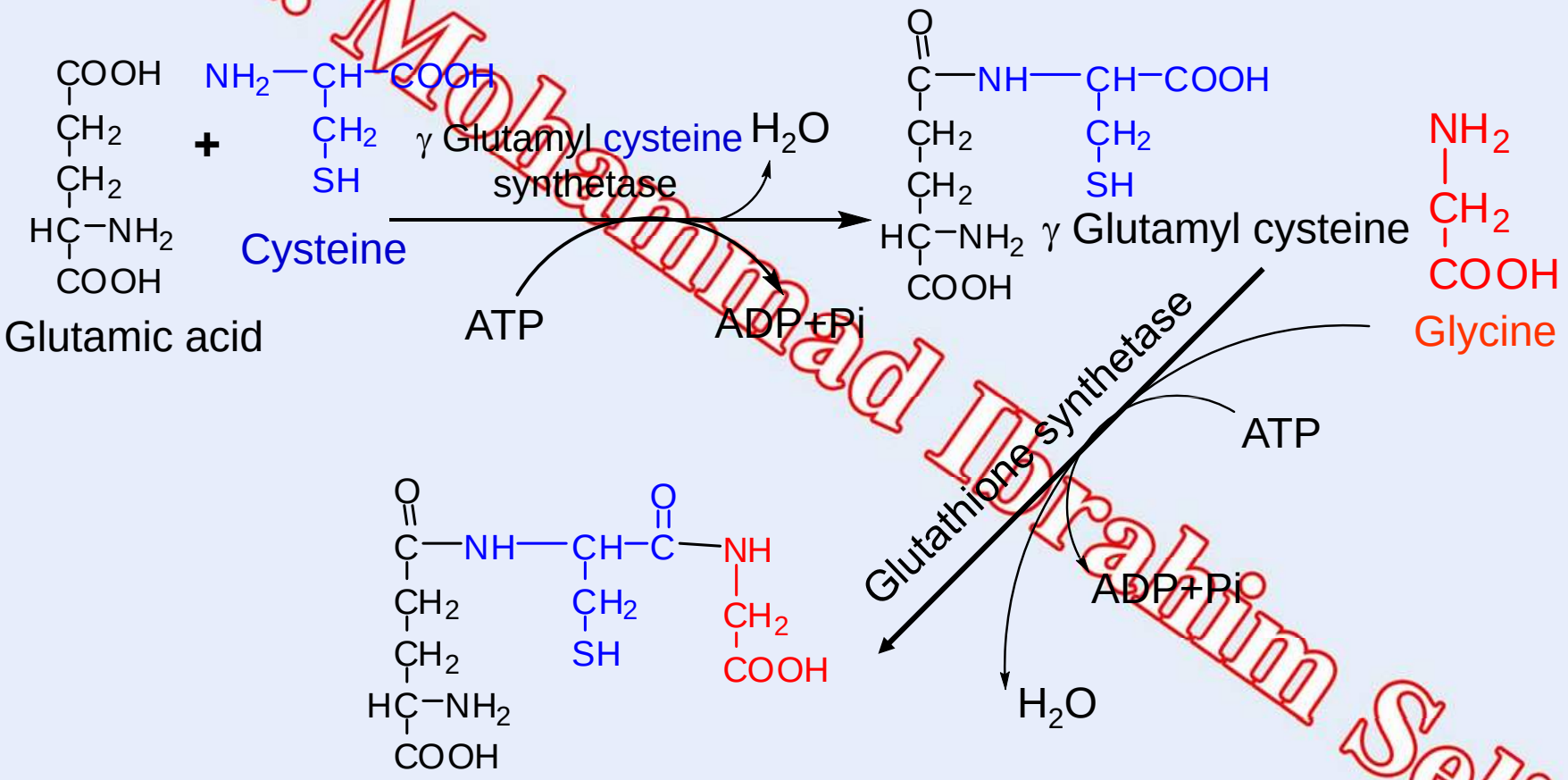
- In children
- In females: during pregnancy and after labour
- Prolonged administration of androgens

Pathological causes

- Degenerative Muscular dystrophy.
  - Myasthenia gravis.
  - Acute poliomyelitis.
- Hypogonadism
- Increase tissue catabolism
  - Hyperthyroidism
  - Starvation
  - Diabetes mellitus
  - Tumors

Glycine

7- Biosynthesis of glutathione



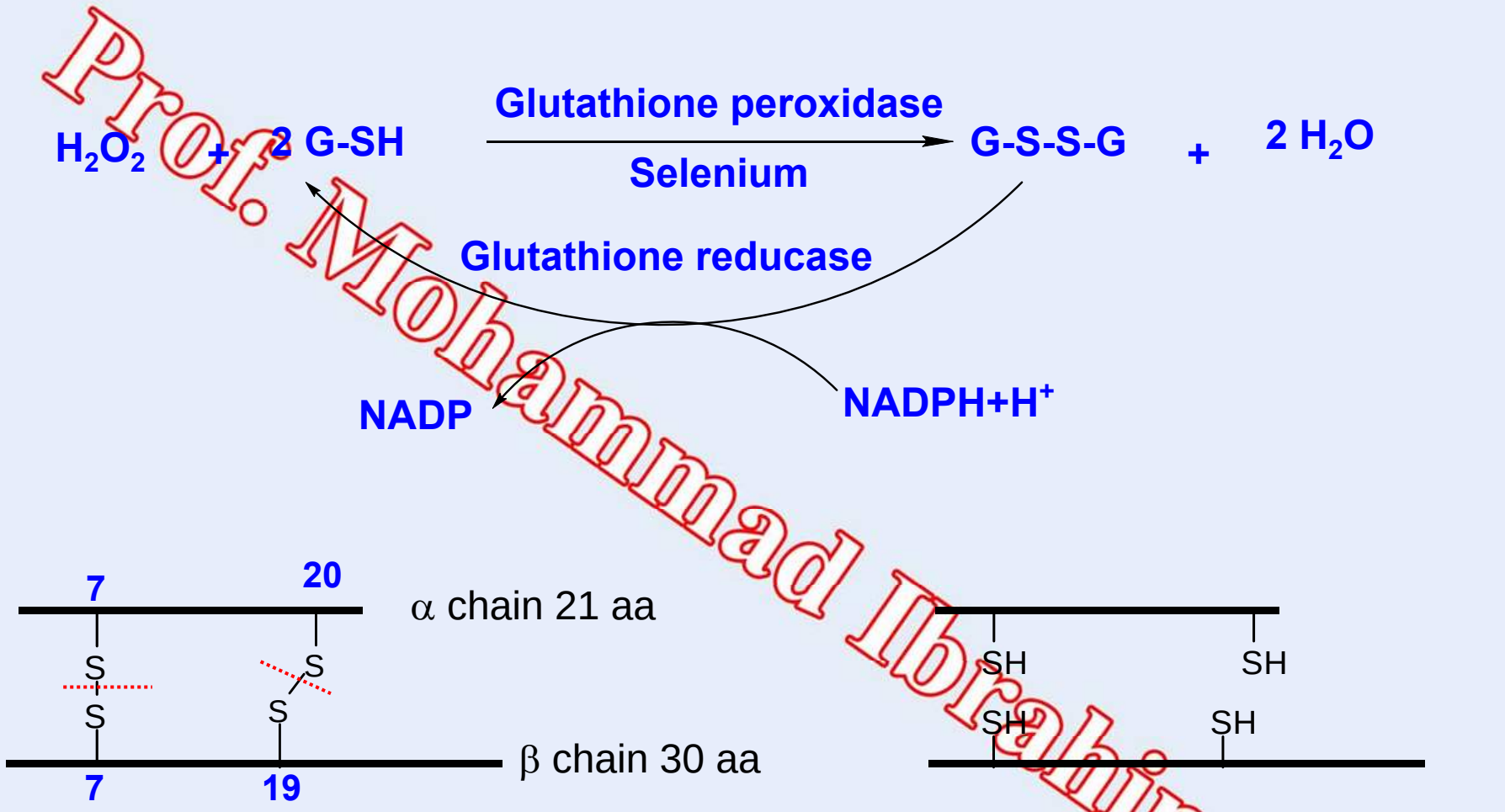
Glutathione (γ Glutamyl cysteinyl glycine)

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Functions of glutathione

- Glutathione protects red cells from haemolysis by H_2O_2
- It protects haemoglobin against oxidation by H_2O_2
- Glutatione has a role in an amino acid transport system
- It inactivates insulin in the liver.
- It maintains (-SH) group, which is important for the activity of many enzymes.
- It protects the B-cells of pancreas from the degenerative action of alloxan
- Glutathione acts as a hydrogen carrier

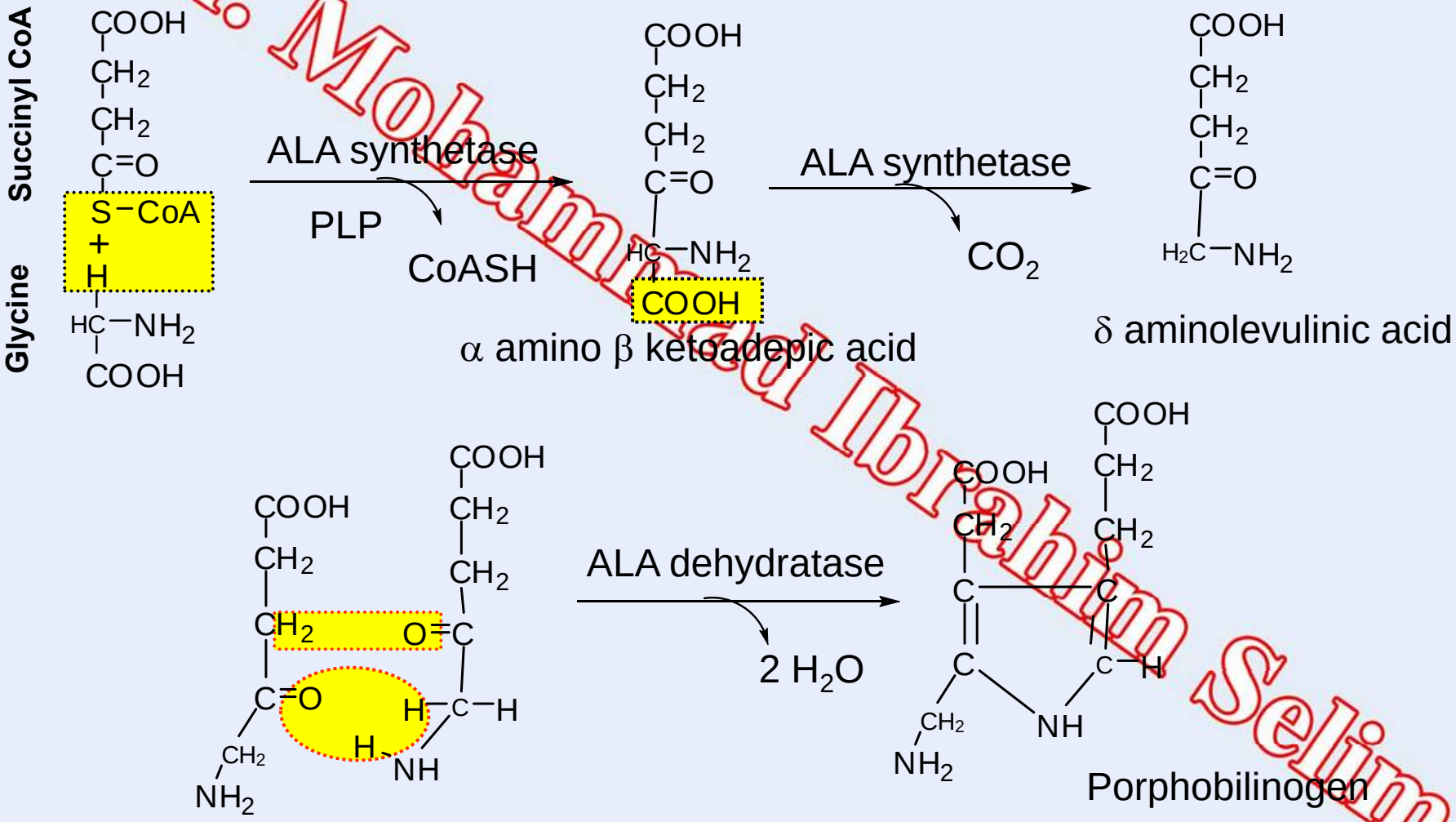
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Glycine

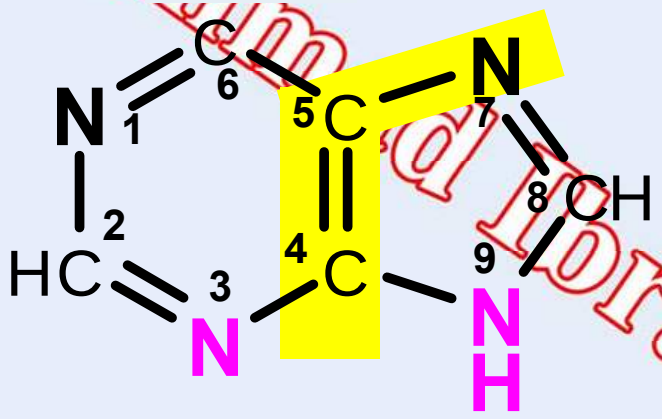
7- Biosynthesis of Haeme



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Glycine

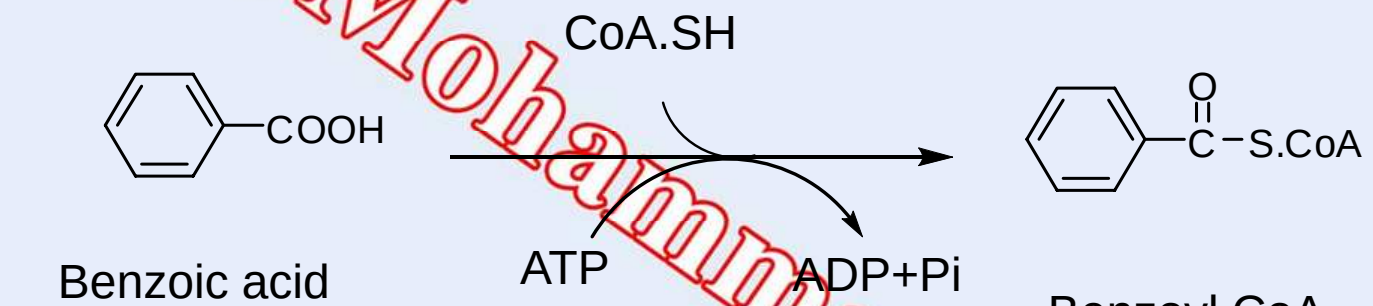
8- Biosynthesis of purine bases

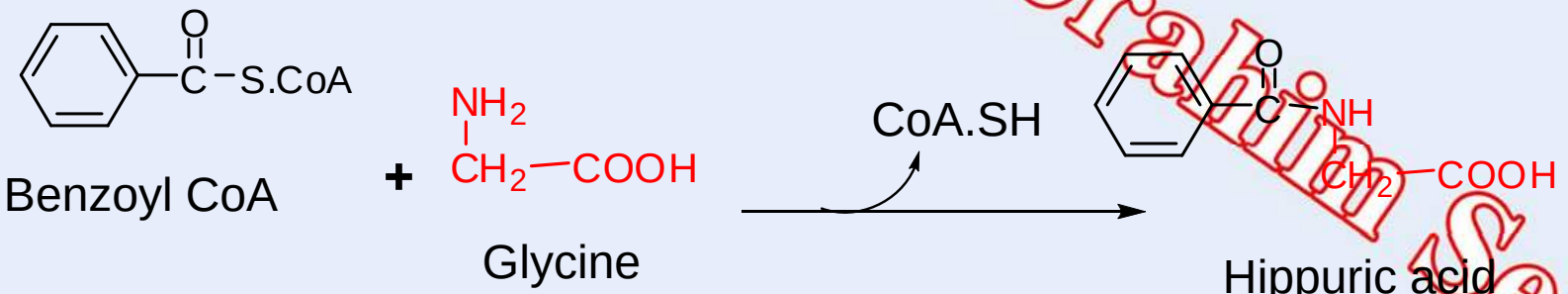


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Glycine

9- Detoxication by conjugation

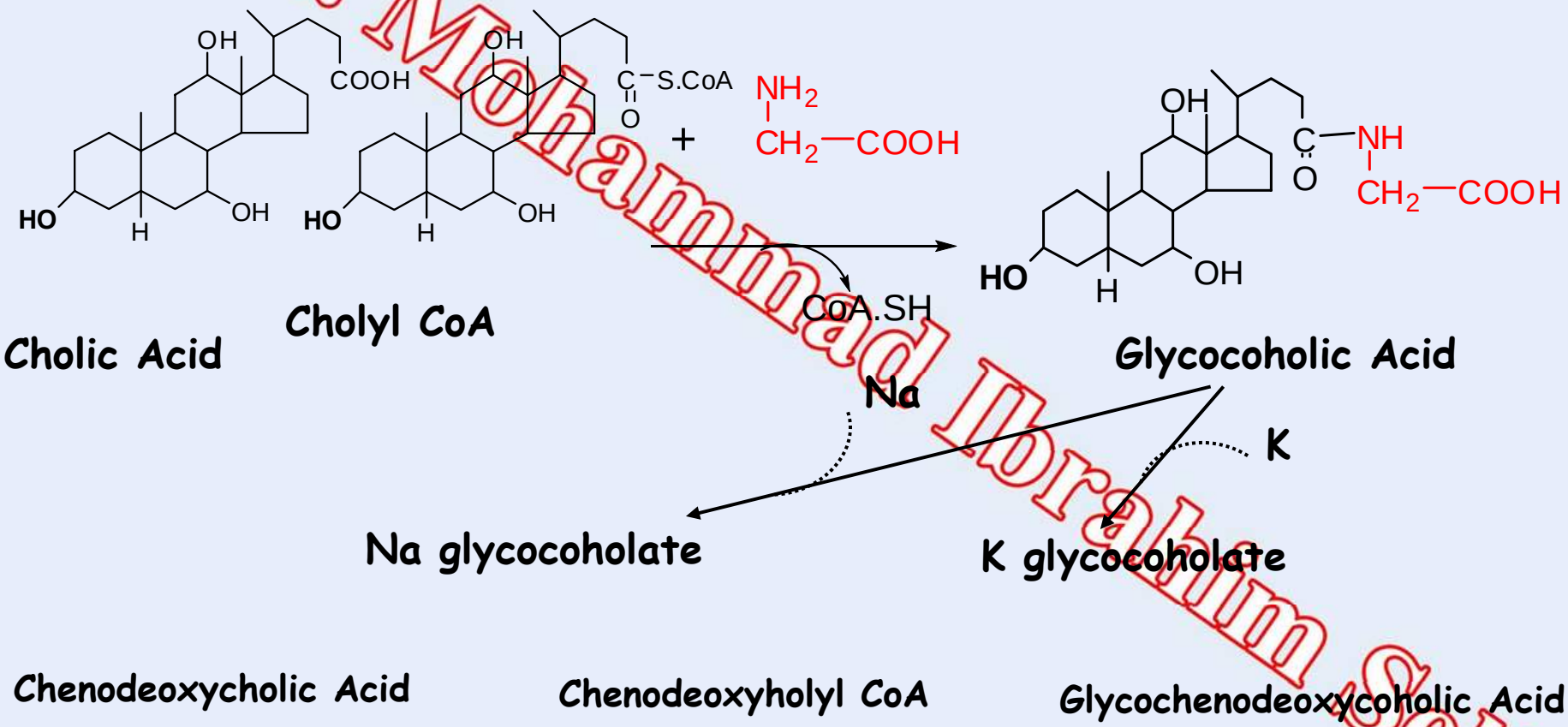




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Glycine

10 – Conjugation with bile acids



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Glycine

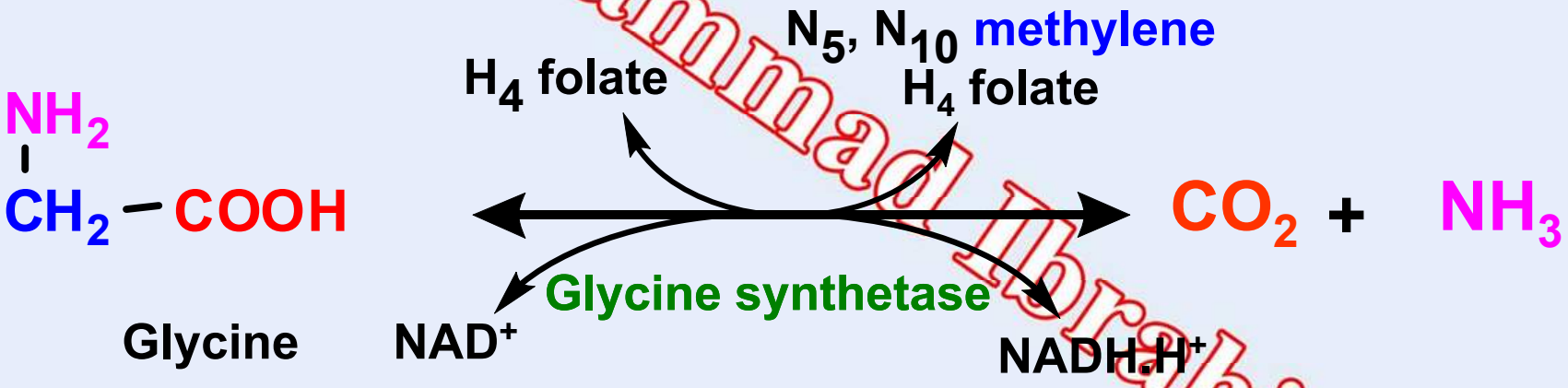
Catabolic Fate

1. Cleavage to CO_2 and NH_3
2. Formation of glyoxylic acid
3. Glycine threonine cycle
4. Glycine succinate cycle

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Glycine

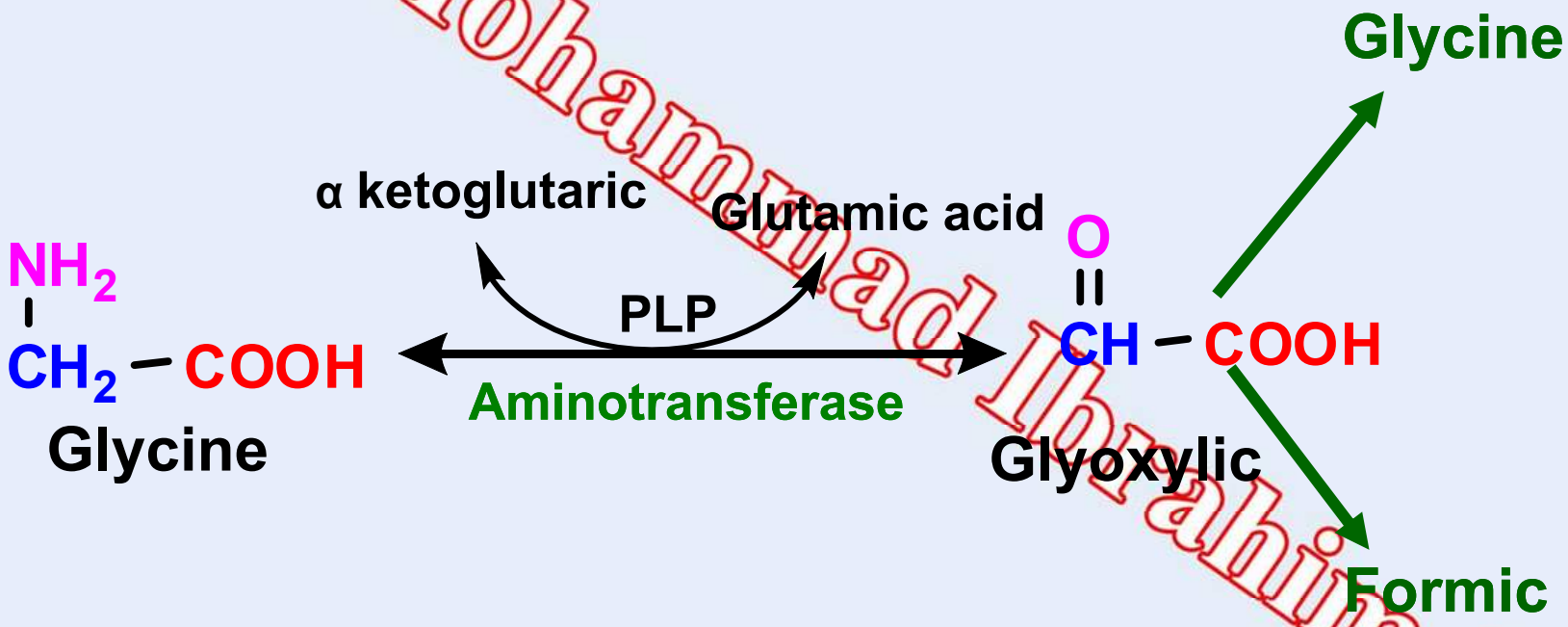
1- Cleavage into CO_2 and NH_3



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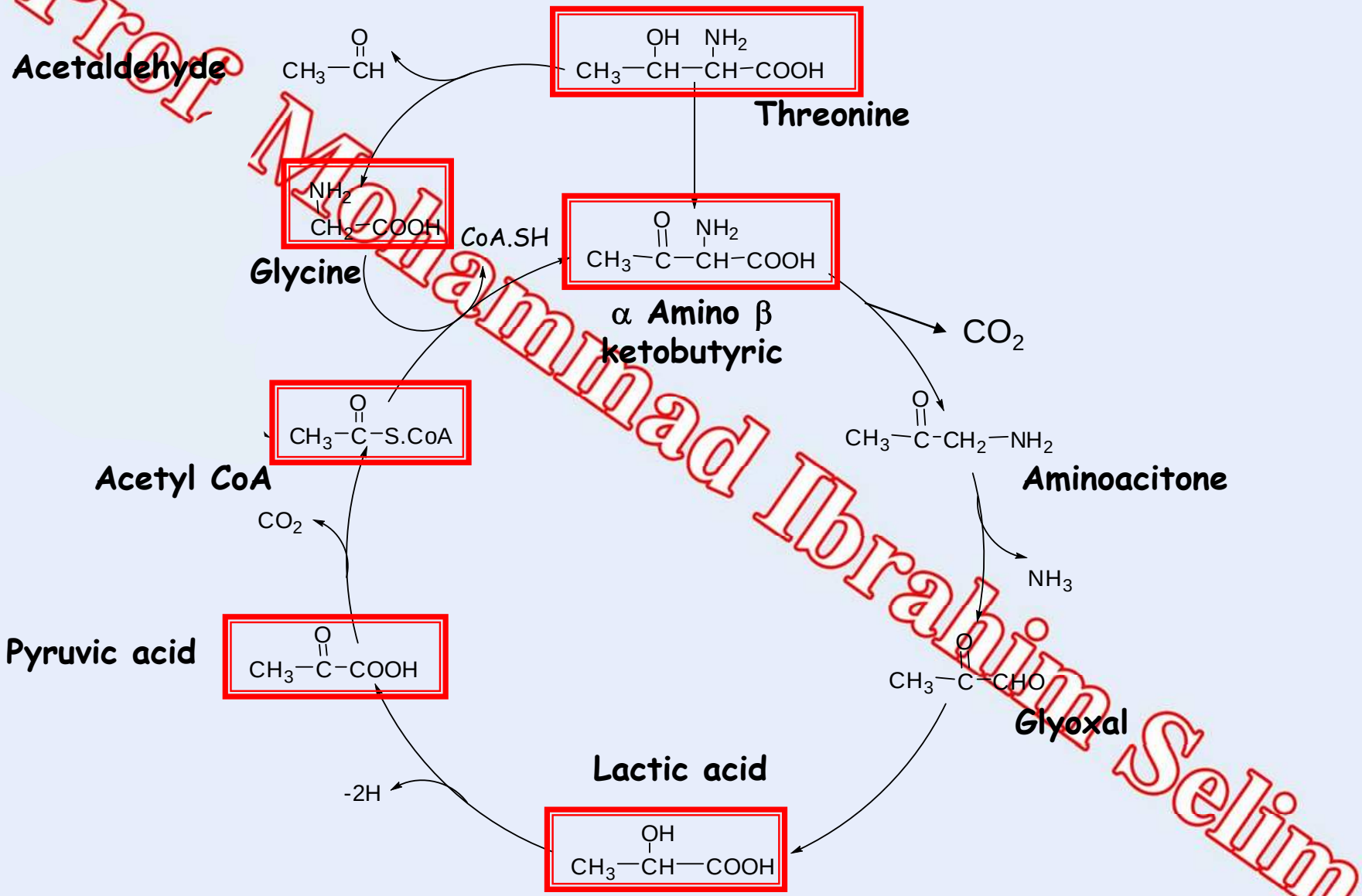
Glycine

2- Formation of glyoxylic acid



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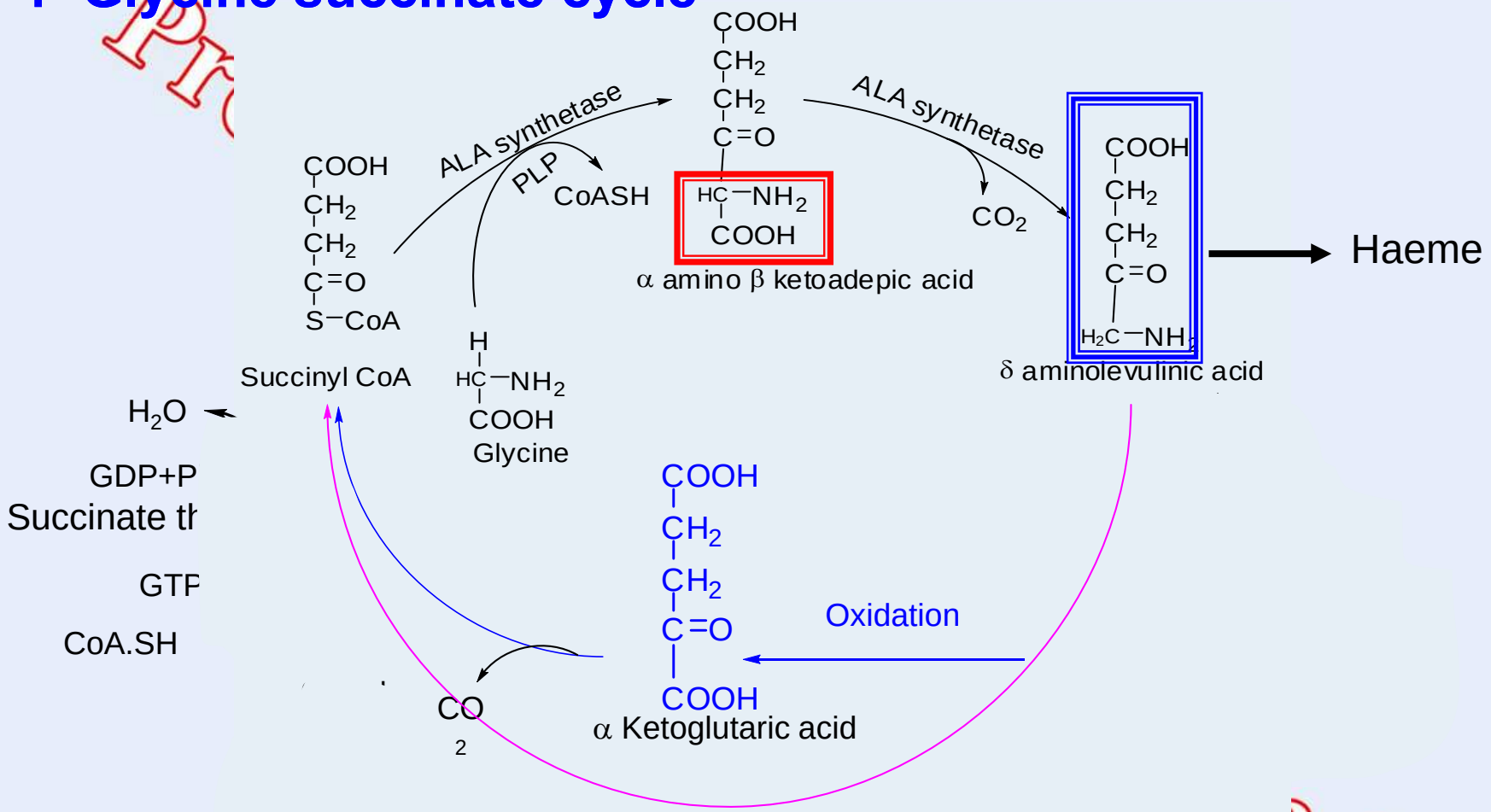
3- Glycine threonine cycle



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4- Glycine succinate cycle



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Glycine

Inborn errors of glycine metabolism

1. Primary hyperoxaluria
2. Glycinuria

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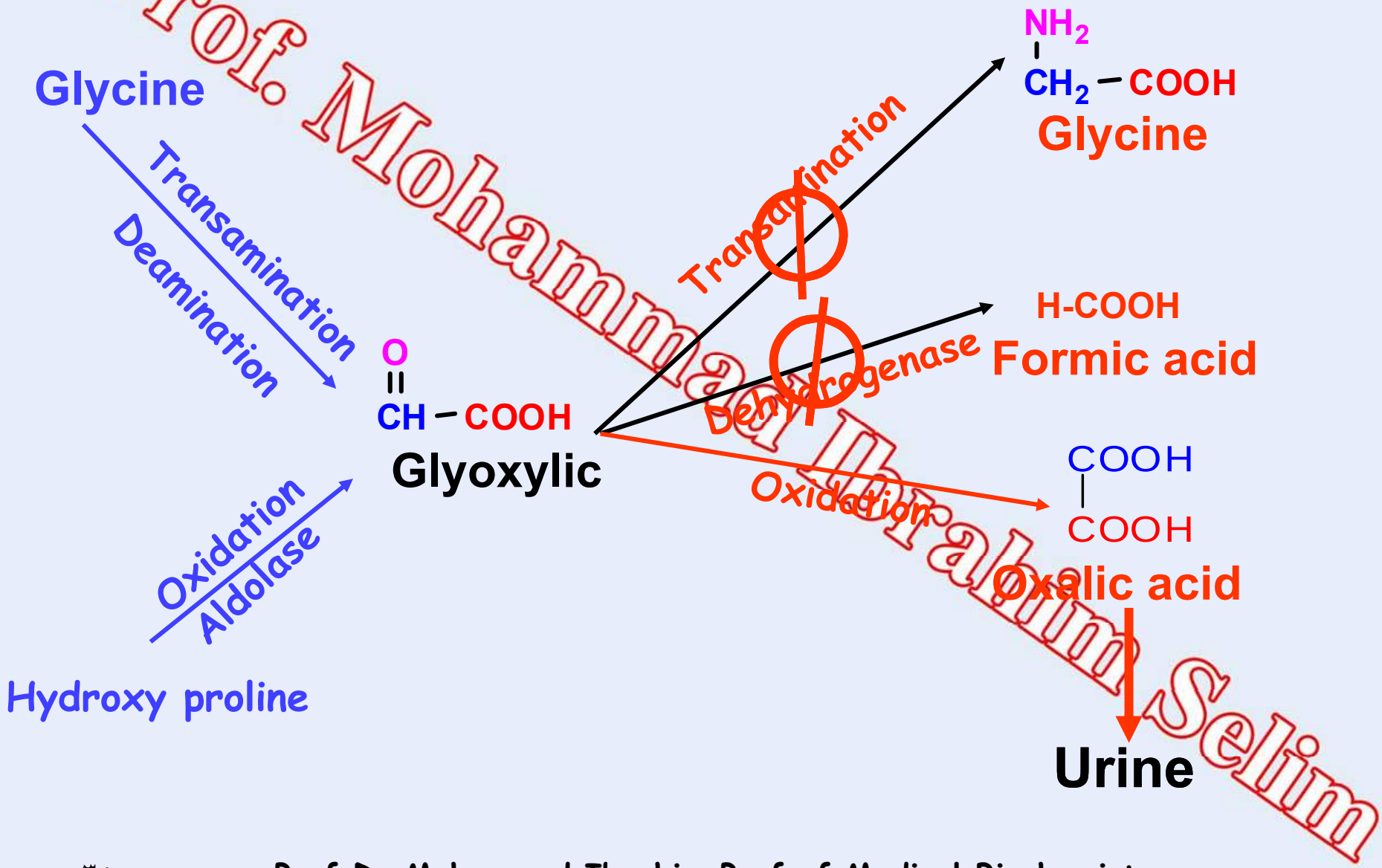
Glycine

Inborn errors of glycine metabolism

1. Primary hyperoxaluria
   - Failure to metabolize glyoxylic acid normally

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Sources and fate of glyoxylic acid



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Glycine

Inborn errors of glycine metabolism

1. Primary hyperoxaluria

- Failure to catabolize glyoxylic acid normally
- Subsequent oxidation of glyoxylic to oxalic acid
- Increased excretion of oxalic acid in urine in the form of calcium oxalate

- This results in:
  - Urolithiasis
  - Nephrocalcinosis
  - Early mortality from
    - Renal failure
    - Hypertension.

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Glycine

Inborn errors of glycine metabolism

2. Glycinuria

- Excess excretion of glycine in urine
- Glycine level in blood is normal
- A defect in renal tubular reabsorption of glycine
- Oxalate excretion in urine is normal.
- Tendency of formation of oxalate urinary stones

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